

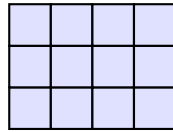
Square or Cubic? Finding the Unit Mistake

Grade 4–5

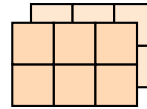
Reading the unit off the formula, finding and fixing a squared-for-cubed mistake, and converting square and cubic units

Directions:

- Every answer needs a **number** *and* the **right kind of unit**. Count how many lengths you multiplied: two lengths give *square* units, three lengths give *cubic* units.
- On a **find the mistake** problem, name the mistake in words first, then show correct work and give the correct answer.
- Write units in the answer blank exactly as shown.



Two lengths multiplied → solid blocks
square units



Three lengths multiplied → solid blocks
cubic units

How units are named. No measurements are shown here — use the numbers given in each problem.

1. A sticker is 8 centimeters long and 5 centimeters wide. Find the area of the sticker.

Answer: _____ cm^2

2. A gift box is 8 centimeters long, 5 centimeters wide, and 3 centimeters tall. Find the volume of the box.

Answer: _____ cm^3

3. Find the mistake. Dev found the volume of a solid cube whose edges are 6 centimeters long. He wrote:

$$V = 6 \times 6 = 36 \text{ cubic centimeters}$$

(a) Name Dev's mistake in words. (b) Show correct work and give the correct volume.

Answer: _____ cm^3

4. A number cube has edges 4 centimeters long. Find its surface area, which is the total area of all six faces.

Answer: _____ cm^2

5. One centimeter is the same length as 10 millimeters. How many square millimeters exactly cover one square centimeter? Draw the square and its rows of tiny squares to prove your answer.

Answer: _____ mm^2

6. Find the mistake. Ana wrote: “*One cubic centimeter is 10 cubic millimeters, because one centimeter is 10 millimeters.*” Her partner said, “*No, it is 100 cubic millimeters.*” **Both** are wrong.

(a) Explain what each of them did with the factor. (b) Give the correct number of cubic millimeters in one cubic centimeter.

Answer: _____ mm^3

7. Find the mistake. A rectangular pool is 12 meters long, 5 meters wide, and 2 meters deep. Kai needs the area of the pool *floor* so he can order a cover. He wrote:

$$12 \times 5 \times 2 = 120 \text{ square meters}$$

(a) Name the two things wrong with Kai's work — what he multiplied, and what he called the answer. (b) Give the correct floor area.

Answer: _____ m²

8. A classmate says: "*Area and volume are the same thing. You just multiply the sides.*" Draw a flat rectangle covered with unit squares, and a box filled with unit cubes. Then write two or three sentences explaining why multiplying two lengths gives square units but multiplying three lengths gives cubic units.